Software Test Plan (STP)

Online Exam Portal

Version: 2.0

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Status: Design Document — Planned Testing Strategy (Pre-Execution)

1. Introduction

Purpose: This document defines the planned test strategy for the Online Exam Portal, covering objectives, scope, approach, resources, and responsibilities for verifying the system against the Software Requirements Specification (SRS v1.0, the authoritative source of truth) and the companion Software Architecture and Design (SAD) Specification. This is a planning document, written before test execution; it defines what will be tested and how, not the results of testing.

Scope: Testing covers Admin and Student web interactions — authentication/RBAC (incl. login lockout and admin password reset), question-bank management with search/filter, exam creation/scheduling with batch assignment, timed exam attempts (auto-save, server-enforced timing, auto-submit, response locking), automatic evaluation with negative marking, gated result visibility and reporting/analytics, bulk student import and batch management, proctoring-lite logging (tab-switch/blur/fullscreen-exit) with admin disqualify/re-permit, and notifications. Excluded: descriptive-question grading, live webcam proctoring, plagiarism detection, and third-party ERP/payment integration (out of scope per SRS §1.2).

References: Online_Exam_Portal_SRS_Final v1.0 (authoritative source, including its Requirements Traceability Matrix at SRS §9); Software Architecture and Design Specification v2.0 (companion document).

Definitions: MCQ, RBAC, JWT, NFR, FR, RTM, TLS, CSV, WCAG — as defined in the SRS §1.4.

2. Test Items

- Auth & RBAC Module (login, lockout, RBAC enforcement, admin password reset, student bulk import, batch management)
- Question Bank Module (CRUD, search/filter by subject/topic/difficulty)
- Exam Management Module (exam creation incl. result_visible_from, question attachment, batch assignment)
- Exam Engine (exam-window visibility, attempt start, autosave, timer, auto-submit, response locking)
- Evaluation Engine (auto-scoring, negative marking, Result creation)
- Reporting Module (gated result retrieval, class/question-wise analytics, export)
- Proctoring-lite Module (tab-switch/blur/fullscreen-exit logging, flagged-attempts view, disqualify/re-permit)
- Notification Module (email on scheduling and on result-visible)

3. Features to be Tested

Every requirement formally defined in the SRS's own Requirements Traceability Matrix (SRS §9) is covered below. Full per-requirement test-case and RTM mapping is in the companion Test Cases & RTM workbook and Coverage Report.

3.1 Functional Requirements (OEP-F-001 through OEP-F-022)

Req. ID	Feature
OEP-F-001	Login with role-scoped session/JWT
OEP-F-002	Login lockout after 5 failed attempts for 15 minutes
OEP-F-003	RBAC enforced at the API level
OEP-F-004	Admin resets a student's password (1-hour, single-use link)
OEP-F-005	Question bank CRUD (2–6 options, exactly one correct)

Req. ID	Feature
OEP-F-006	Search/filter questions by subject, topic, difficulty
OEP-F-007	Answer key never exposed to Student clients
OEP-F-008	Create exam incl. result_visible_from
OEP-F-009	Attach questions to exam (manual/random)
OEP-F-010	Assign exam to student batch(es)
OEP-F-011	Bulk student import (CSV/Excel)
OEP-F-012	Batch/section creation and student mapping
OEP-F-013	Exam visibility limited to assigned batch and scheduled window
OEP-F-014	Single active attempt per (exam, student)
OEP-F-015	Auto-save every 10 seconds
OEP-F-016	Server-side auto-submit on timeout
OEP-F-017	Auto-evaluation with negative marking
OEP-F-018	Response locking once attempt leaves IN_PROGRESS
OEP-F-019	Class/question-wise analytics reporting
OEP-F-020	Tab-switch/blur proctoring log and flagged-attempts view
OEP-F-021	Admin disqualify/re-permit with audit log
OEP-F-022	Email notification on scheduling and result-visible

3.2 Non-Functional Requirements (OEP-NF-001 through OEP-NF-006)

Req. ID	Feature
OEP-NF-001	Attempt page load ≤2s; ≥200 concurrent students/exam, ≤3s p95 degradation
OEP-NF-002	Horizontal scalability; DB indexed on (exam_id, student_id)
OEP-NF-003	99.5% availability during published exam windows
OEP-NF-004	Auto-save ≤10s apart; crash loses ≤10s of activity
OEP-NF-005	Responsive UI; WCAG 2.1 AA basics on exam-taking screens
OEP-NF-006	≥70% automated unit-test coverage on core modules

3.3 Security Requirements (OEP-SR-001 through OEP-SR-007)

Req. ID	Feature
OEP-SR-001	Salted password hashing; no plaintext storage/logging
OEP-SR-002	HTTPS/TLS 1.2+; HttpOnly + Secure session cookies
OEP-SR-003	Correct-option flag never sent to Student clients
OEP-SR-004	RBAC enforced server-side on every endpoint
OEP-SR-005	Single active session per (exam, student) attempt
OEP-SR-006	Append-only audit log of admin overrides
OEP-SR-007	Input validation / parameterized queries against injection and stored XSS

4. Features Not to be Tested

- Descriptive/subjective question grading (excluded from SRS scope, §1.2)
- Live webcam proctoring and plagiarism detection (excluded from SRS scope, §1.2)
- External university ERP / payment gateway integration (excluded from SRS scope, §1.2)
- Third-party SMTP provider's internal delivery infrastructure (assumed reliable; only the outbound call is tested)

5. Test Approach / Strategy

5.1 Test Levels

- Unit Testing: individual functions/serializers/permission classes (e.g., scoring logic, answer-key exclusion, RBAC checks, lockout logic) using Pytest/Django test client.
- Integration Testing: module-to-module and module-to-database interactions (e.g., Exam Management → Question Bank → Notification).
- System Testing: end-to-end flows through the API and UI (e.g., full student exam attempt from login to gated result view).
- Acceptance Testing: verification against each requirement's SRS acceptance criteria (SRS §4–§5.1.2) by the course instructor/evaluator.

5.2 Test Types

- Functional testing of all core features listed in Section 3.1.
- Security testing: answer-key confidentiality, password hashing, TLS/cookie flags, single-active-attempt enforcement, RBAC boundaries, audit logging, input validation/injection checks (Section 3.3).
- Performance testing: response time and concurrent-attempt load consistent with OEP-NF-001/002.
- Regression testing after each significant change to shared modules (Auth, Exam Engine).
- Usability/accessibility testing: timer visibility, WCAG 2.1 AA basics (OEP-NF-005).

5.3 Entry and Exit Criteria

Entry Criteria: a stable build is deployed to the test environment; test data (question bank, sample batches/students) is seeded; test cases are reviewed and approved.

Exit Criteria: 100% of planned test cases executed; all High-priority test cases passed; zero open critical/security defects; the RTM shows every SRS requirement (OEP-F-001..022, OEP-NF-001..006, OEP-SR-001..007) covered by at least one passed test case or a documented operational-verification method.

5.4 Security Testing Detail

- Verify Option.is_correct is absent from every Student-reachable API response (static review of serializers + dynamic API inspection) — OEP-F-007/OEP-SR-003.
- Verify password hashes are salted/adaptive and never appear in logs or the plaintext DB — OEP-SR-001.
- Verify all traffic is served over TLS 1.2+ and session cookies carry HttpOnly and Secure flags — OEP-SR-002.
- Attempt concurrent session creation for the same (student, exam) pair and confirm rejection — OEP-F-014/OEP-SR-005.
- Attempt admin-only endpoints with a Student token and confirm 403 on every one — OEP-F-003/OEP-SR-004.

- Confirm a login is locked for 15 minutes after 5 consecutive failed attempts, and the event is logged — OEP-F-002.
- Confirm every disqualify/re-permit action creates an immutable AuditLog entry with actor and timestamp — OEP-SR-006/OEP-F-021.
- Fuzz login and question-bank input fields (OWASP ZAP / manual payloads) for injection and stored-XSS handling — OEP-SR-007.

6. Test Environment

Category	Details
Application	Online Exam Portal backend (Django/DRF) + responsive web client, deployed via Docker on a Linux VM
Database	PostgreSQL 14+ (isolated test schema/instance, seeded with fixture data)
Cache	Redis (test instance, optional per SRS)
Tools	Pytest / Django test client (unit & integration), Postman/Newman (API), Selenium (UI/system), JMeter or Locust (performance), OWASP ZAP (security), Lighthouse (accessibility), Git-based issue tracker for defects
Test Data	Synthetic Admin and Student accounts, a seeded question bank (≥ 50 questions across subjects/topics/difficulties), 2–3 sample batches

7. Test Schedule

Milestone	Target
Test case design & review	Week 1 of the testing phase
Test environment setup & data seeding	Week 1
Unit & integration test execution	Week 2
System & security test execution	Week 3
Performance & accessibility test execution	Week 3
Regression pass & defect closure	Week 4
Acceptance review with instructor/evaluator	Week 4

8. Test Deliverables

- This Software Test Plan
- Test Cases document (companion spreadsheet, covering all OEP-F/NF/SR requirements)
- Requirements Traceability Matrix (companion spreadsheet)
- Test execution logs and defect reports (produced during the execution phase, once a build exists)
- Test Summary Report (produced at the end of the execution phase)

9. Roles and Responsibilities

Role	Name	Responsibility
Developer / Test Lead	Chiranthan V	Own the test plan; coordinate execution and reporting

Role	Name	Responsibility
Developer / Tester	Dwijesh Krishna Ghattamaraju	Design & execute unit/integration test cases; log defects
Developer / Tester	Harkirat Singh	Design & execute system/security test cases; log defects
Course Instructor / Evaluator	Dr. Pradeep Kumar Kenny S	Review deliverables; approve acceptance criteria

10. Risks and Mitigation

Risk	Mitigation
Implementation delayed relative to the semester schedule	Prioritize High-priority requirements (auth, exam engine, evaluation) for earliest test coverage
Timer/auto-submit logic is difficult to test deterministically	Use a mockable server clock in the test environment to simulate timeout without real-time waits
result_visible_from gating is time-dependent and easy to under-test	Use a mockable server clock to test both sides of the boundary (just before / at / after result_visible_from)
Limited team bandwidth (3-person team) for parallel dev + test	Automate regression-prone areas (auth, scoring) first; keep manual testing focused on UI/UX flows
Test environment instability (shared VM)	Maintain a docker-compose based local fallback environment

11. Assumptions and Dependencies

- A working build implementing at least the High-priority requirements (Section 3.1) will be available before system/security testing begins.
- Test data (question bank, sample rosters) will be prepared ahead of the execution phase.
- The test environment mirrors the production technology stack (SAD §3.6).

12. Suspension and Resumption Criteria

Suspend testing if: the test environment is unavailable for more than 4 hours, or the build is unstable enough to block more than 30% of planned test cases.

Resume testing when: blocking defects are resolved and the environment is confirmed stable.

13. Test Case Management and Traceability

Every test case in the companion Test Cases document carries a Requirement_ID that maps back to the SRS. The companion Requirements Traceability Matrix (RTM) consolidates Requirement → Module → Component → API → Test Case mappings for all 35 formally-defined SRS requirements (22 FR + 6 NFR + 7 SR). Representative examples:

Requirement	Test Case(s)
OEP-F-001 (Login/RBAC)	TC-AUTH-01, TC-AUTH-02, TC-AUTHZ-01
OEP-F-002 (Login lockout)	TC-AUTH-04
OEP-F-011 (Bulk student import)	TC-REG-01, TC-REG-02, TC-REG-03
OEP-F-012 (Batch/section management)	TC-BATCH-01

Requirement	Test Case(s)
OEP-F-015 (Auto-save)	TC-TIMER-02, TC-TIMER-03
OEP-F-016 (Auto-submit)	TC-TIMER-01
OEP-F-017 (Auto-evaluation)	TC-RESULT-01, TC-RESULT-02
OEP-F-018 (Response locking)	TC-SEC-06
OEP-F-020 (Proctoring-lite log)	TC-PROC-01
OEP-F-021 (Disqualify/re-permit)	TC-PROC-02
OEP-NF-001 (Performance/concurrency)	TC-PERF-01
OEP-SR-002 (TLS/cookie flags)	TC-SEC-04
OEP-SR-003 (Answer key hidden)	TC-SEC-01
OEP-SR-006 (Audit log / non-repudiation)	TC-SEC-05
OEP-SR-007 (Injection/XSS hardening)	TC-SEC-03

14. Test Metrics and Reporting

- % of planned test cases executed and % passed/failed, tracked weekly during the execution phase.
- Defect density per module and defect aging (time-to-close).
- Requirement coverage percentage, derived from the RTM (target: 100% of the 35 formally-defined SRS requirements).

Reporting cadence: a brief status update at each weekly milestone (Section 7), and a final Test Summary Report at the end of the execution phase.

15. Approvals

Role	Name	Signature / Date
Test Lead	Chiranthan V	
Developer	Dwijesh Krishna Ghattamaraju	
Developer	Harkirat Singh	
Course Instructor / Evaluator	Dr. Pradeep Kumar Kenny S	14-9-2026